2. Problem : The pulverizer

(a) If k|n (316), frog can't reach its food.

(b) number of jump = j

total jump amount : jk

total circle it rotate = C

$$jk = cn + 1$$

$$jk - cn = 1$$

$$jk + (-c)n = 1 \rightarrow (-c)n + jk = 1$$

(c) $n = 50,\ k = 21,\ j = ?$

n	k	(rem(k,n)) = k − q·n
50	21	8 = 50 − 2·21
21	8	5 = 21 − 8·2 = 21 − 2·(50 − 21·2) = −2·50 + 5·21
8	5	3 = 8 − 1·5 = (50 − 2·21) − (−2·50 + 5·21) = 3·50 − 7·21
5	3	2 = 5 − 1·3 = (−2·50 + 5·21) − 1(3·50 − 7·21) = −5·50 + 12·21
3	2	1 = 3 − 1·2 = (3·50 − 7·21) − 1(−5·50 + 12·21) = 8·50 − 19·21
2	1	0

$1 = 8 \cdot 50 - 19 \cdot 21$

$= -(-8)n + (-19)k$

$= -(-8) \cdot 50 + (-19) \cdot 21$

$-21 \cdot 50 + 50 \cdot 21$

$-13 \cdot 50 + 31 \cdot 21$

$= (-13) \cdot 50 + 31k$

we will mody it to get positive j

Problem 2

Pf: by induction

$P(n) ::= \gcd(F_n, F_{n+1}) = 1$

Base case $P(0) = \gcd(0, 1) = 1$ ✓

inductive step: $\forall n \geq 0$, show $P(n) \Rightarrow P(n+1)$

Assume $P(n)$ is true for the purpose of induction

$P(n+1) = \gcd(F_{n+1}, F_{n+2})$

Assume F_{n+1} & F_{n+2} are not relatively prime for the purpose of contradiction $\Rightarrow \gcd(F_{n+1}, F_{n+2}) > 1$

$d > 1$

$d \mid F_{n+2} - F_{n+1}$ (linear combination)

$d \mid F_n$ $(F_{n+2} = F_{n+1} + F_n)$

$d \mid F_n$ & $d \mid F_{n+1}$

$\gcd(F_n, F_{n+1}) = d$

This is contradiction to I.H $P(n)$

$\therefore$ $P(n+1)$ is True

By the principle of induction, $\forall n \geq 0$ $P(n)$ is True.